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Advanced Computer Networks

Inter-domain Routing Security

What happened last lecture?

DNS(SEC) and Email

DNS(SEC)

Service bindings via DNS

The generic DNS record “SVCB” can be instantiated into records specific to different protocols, e.g. HTTPS.

Goal: Provide all required information to make a connection to a service
Fewer DNS queries, so less round-trip time for application protocol connections.

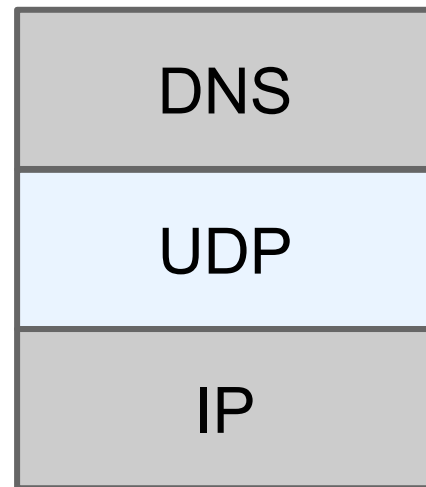
Example:

```
dig www.google.de HTTPS
```

```
;; ANSWER SECTION:
```

```
www.google.de.      21600      IN         HTTPS      1
```

Some problems can be solved



Unreliable

DNS over TCP
(TCP/53)

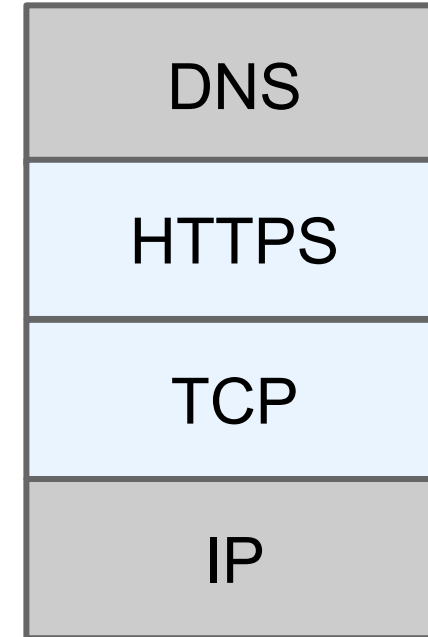
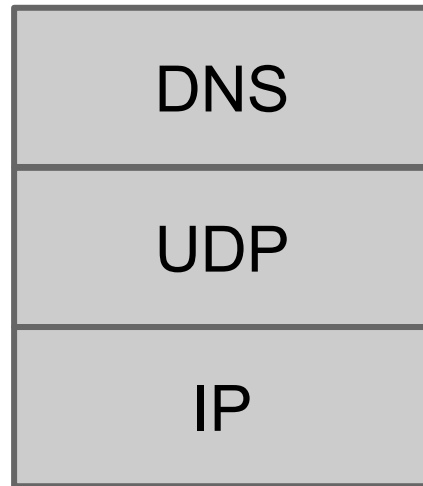
Security/
Privacy

DNS over TLS
(TCP/853)

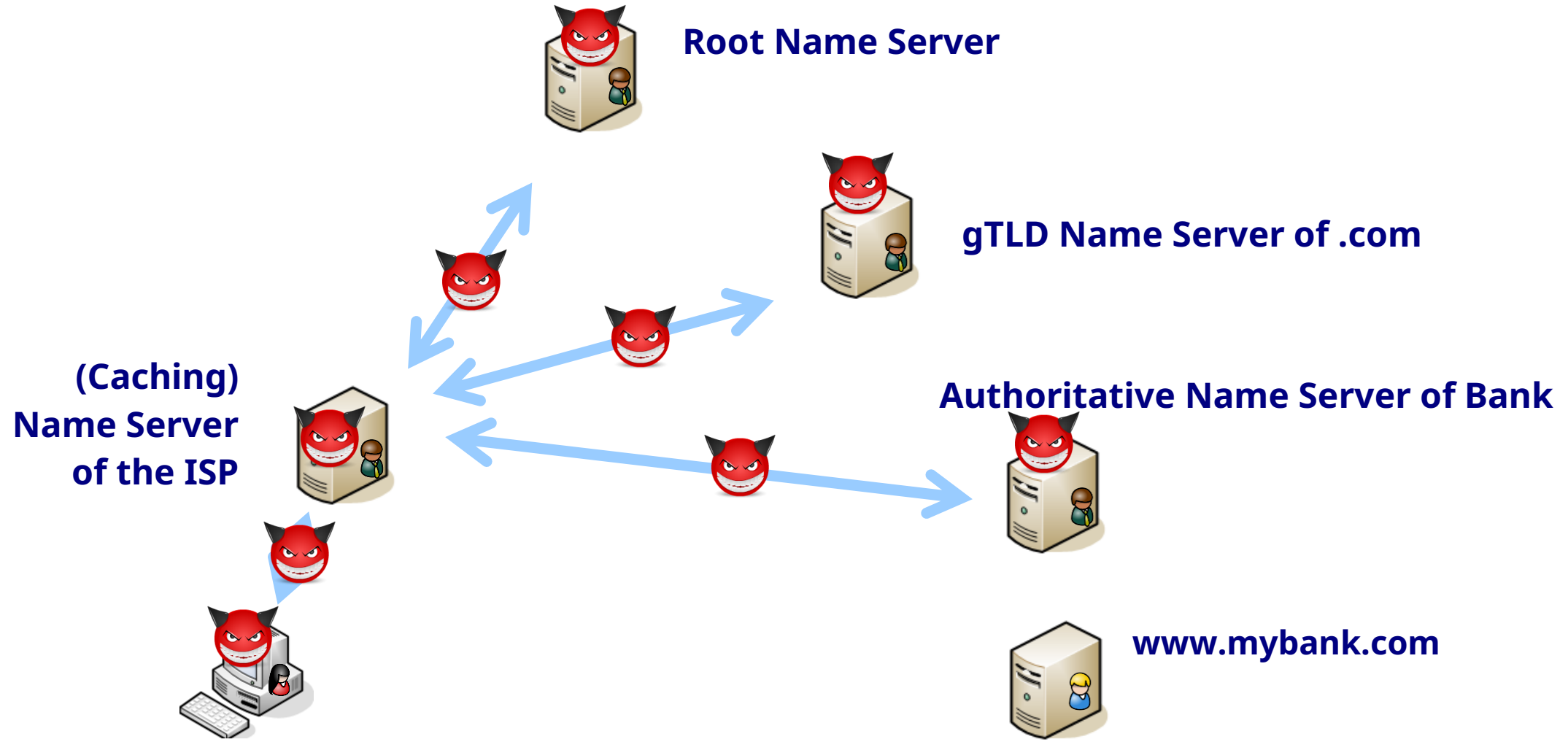
Easy to filter

?

DNS over HTTPS (DoH) [RFC 8484]



Attack surface of DNS



DNSSEC Design Objectives [Current DNSSEC specs: RFCs 4033, 4034, 4035 (orig. 2535)]

Goals: Provide **integrity** to prevent spoofing and poisoning.

Approach:

Authenticating messages of name servers

Check that no one on the way changed the DNS content

Authenticating resource records

Check that data comes originally from authoritative DNS server

Proof of non-existence (prevent DoS against names)

Non-Goals: Confidentiality by hiding DNS data or requests, Authorization of requests or requestors, Protection against DDoS attacks (via traffic amplification)

DNSSEC Fundamentals

DNSSEC is based on public private key cryptography,
and introduces new resource records

RRSIG

contains cryptographic signature

DNSKEY

contains public signing key

DS

contains hash of the DNSKEY

NSEC/NSEC3

for explicit denial-of-existence of DNS records

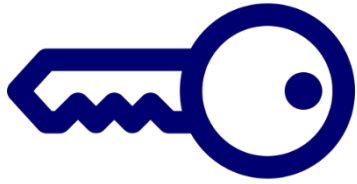
CDNSKEY/CDS

for a child zone requesting updates to DS
record(s) in the parent zone

Secure a zone

Secure zone
delegation

DNSSEC distinguishes between two types of key pairs



Key-Signing-Key

KSK

To validate the ZSK



Zone-Signing Key

ZSK

To sign RRsets in the zone

DNSSEC applies a chain of trust to implement security

Start with a **root zone key**, the highest possible key in the DNS tree

In 2010, the first root full production DNSSEC root zone was signed

Then following cryptographic pointers to lower zones

In November 2011, more than 25% of top-level domains are signed with DNSSEC;

In February 2018, 91% (http://stats.research.icann.org/dns/tld_report/)

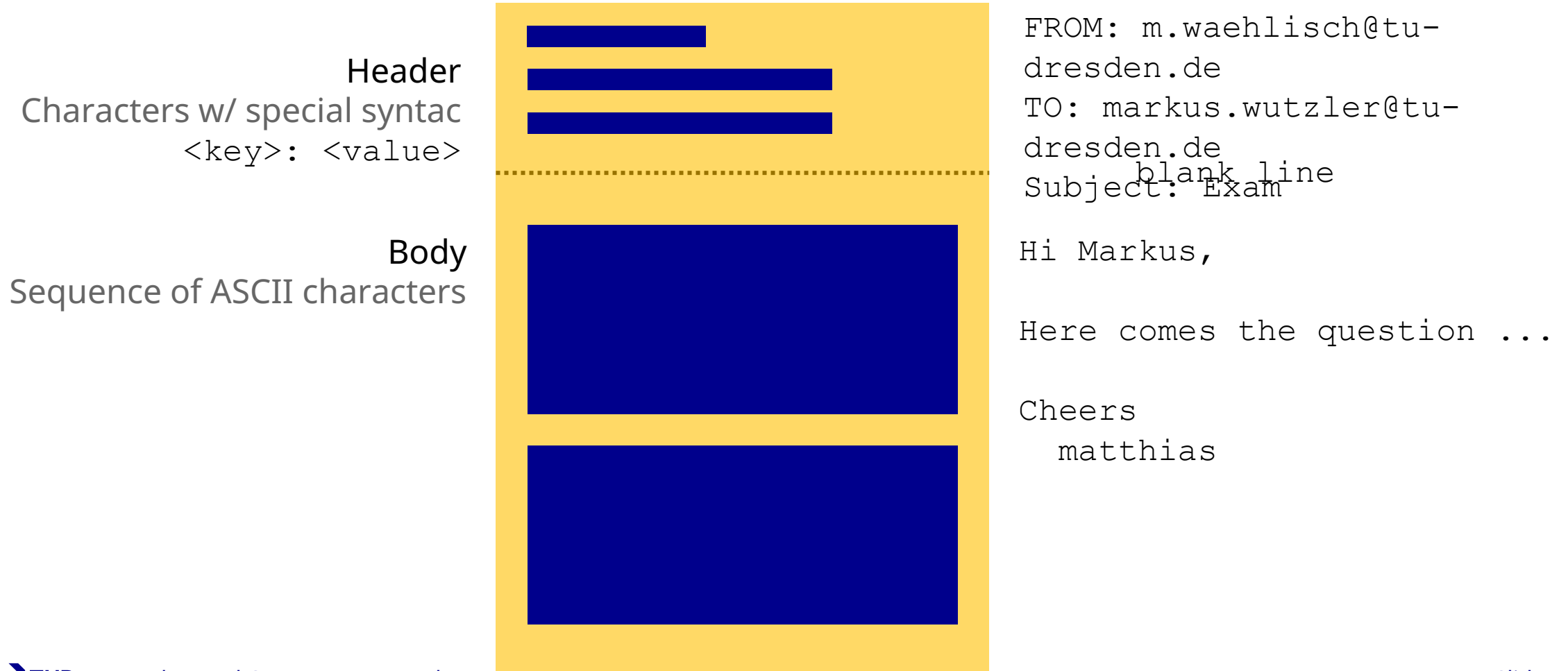
Each pointer is validated with the previous validated zone key

Using this mechanism only the root key is needed to validate *all* DNSSEC keys on the Internet

With these DNSSEC keys the DNS data in each zone can then be validated

Email

Message content consists of Header Fields and Body



High-level overview of email architecture

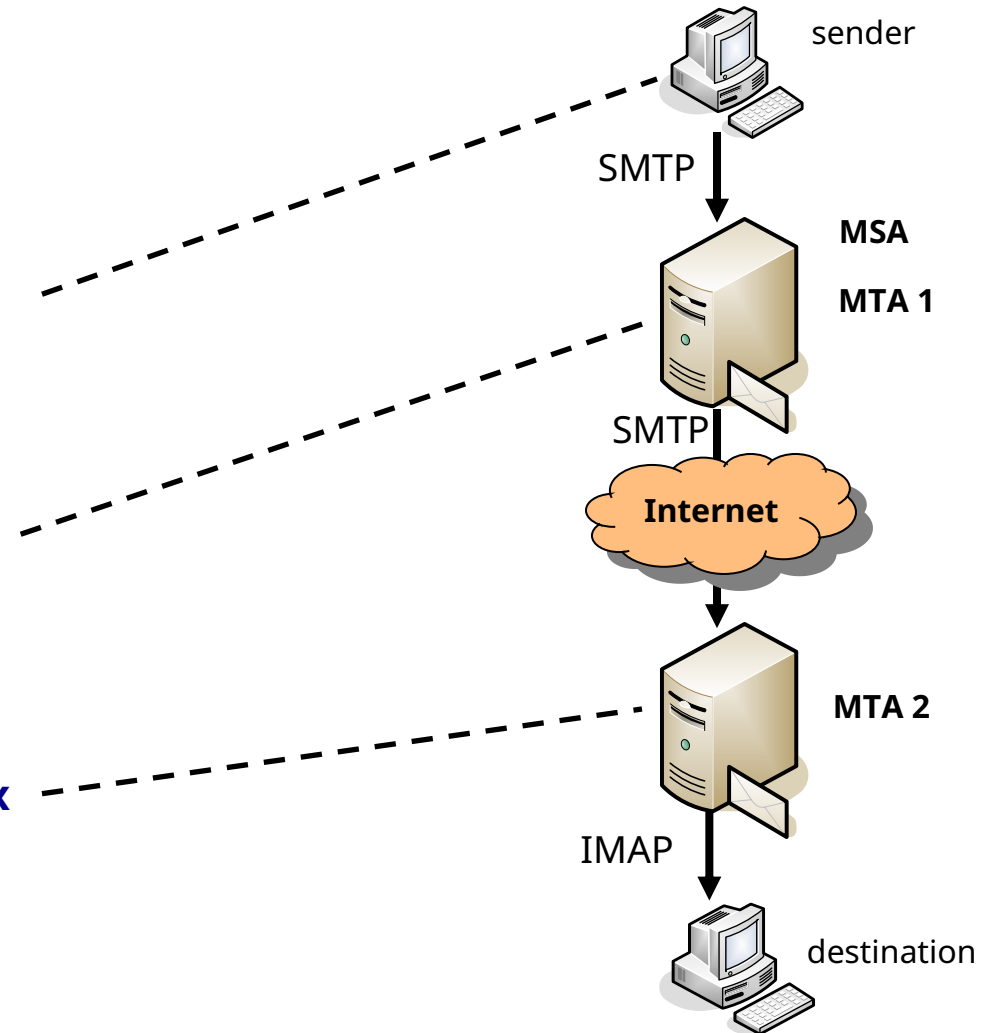
User 1: writes an email

Client 1 (UA 1): formats the email, produces the receiver list, and sends the email to its mail server (MSA of MTA 1, through port 587)

Server 1 (MTA 1): Sets up a connection to the destination's SMTP server (MTA 2) on port 25 and sends a copy of the email

Server 2 (MTA 2): Produces the header of the email and places the email into the destination's mailbox

Users 2: access and read the email, using IMAP, website etc.

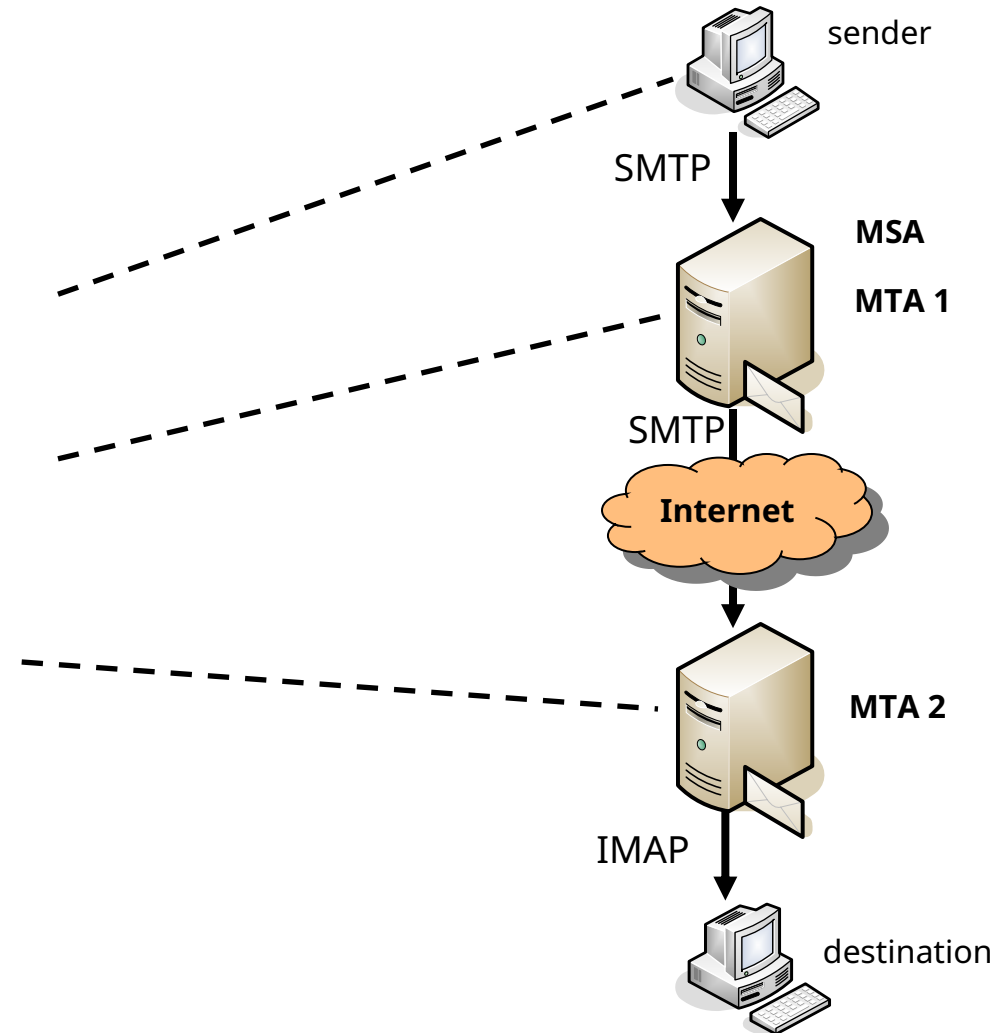


Challenges

Sender may send unencrypted email

Sender may spoof the source email address

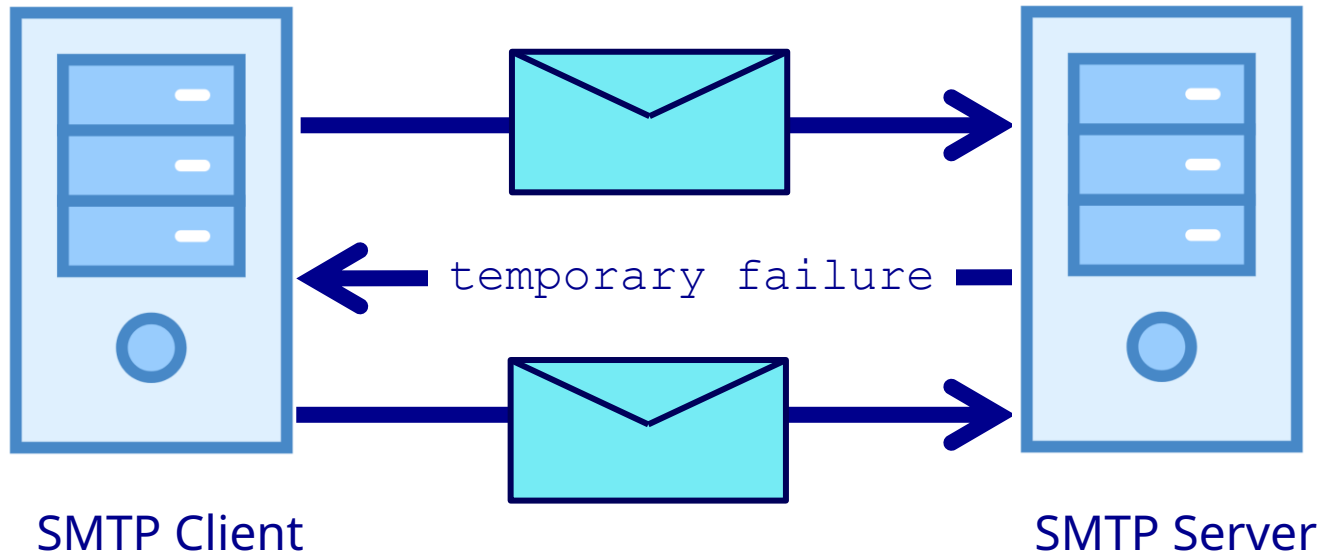
SMTP is not encrypted



Email spoofing, the reason why spam works

Who knows that the source email address is correct?

Mitigate email spoofing using greylisting (RFC 6647)

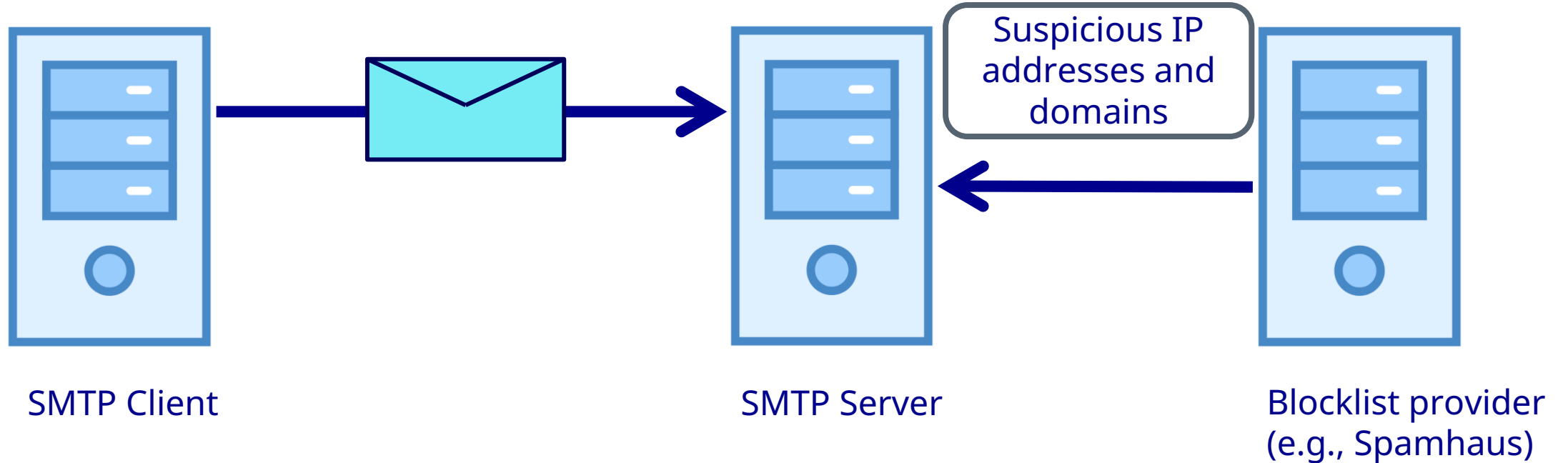


Did I receive an email before from this client?

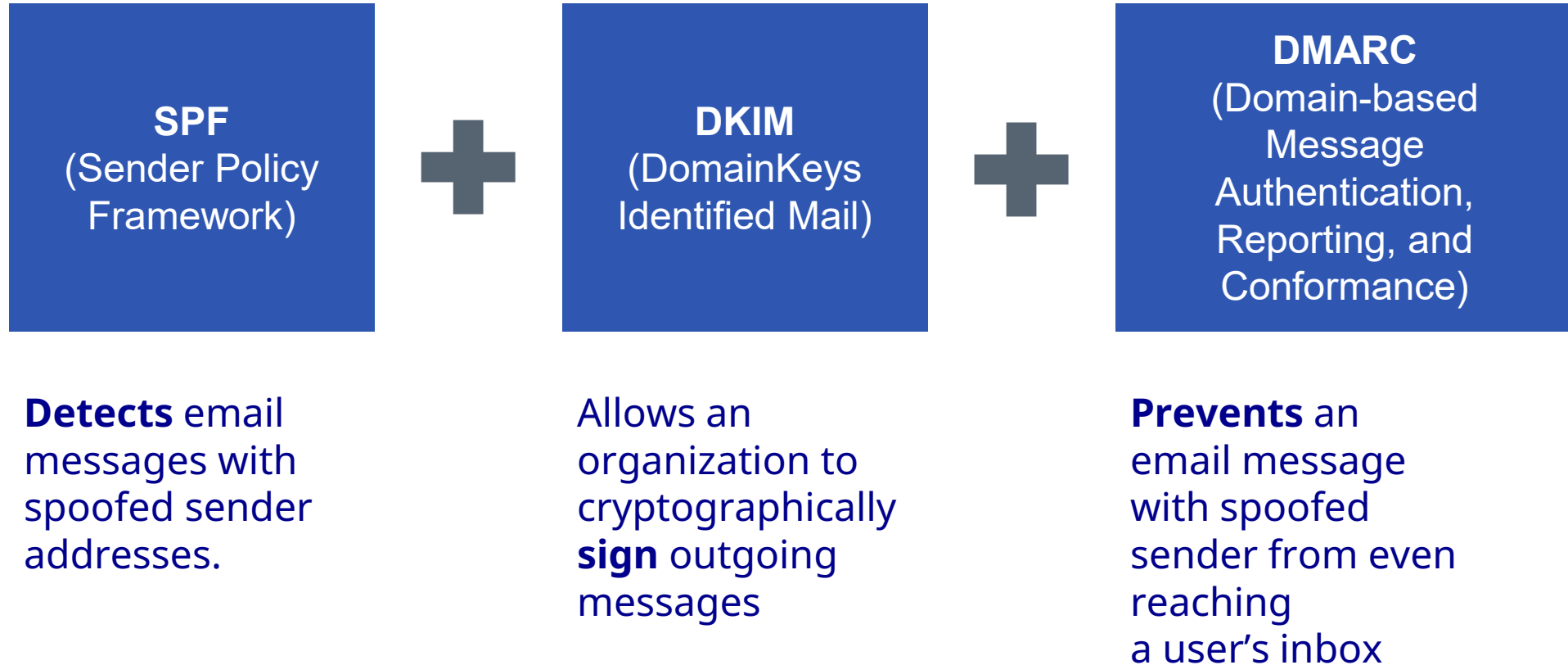
No, then respond with 4xx temporary failure.

Greylisting introduces delays, and email delivery fails if an implementation does not attempt redelivery.

Mitigate email spoofing using blocklisting



Mitigate email spoofing Beyond greylisting



SPF: Sender Policy Framework (RFC 7208)

An SPF policy is used to designate the systems that are authorized to send email on behalf of a domain.

A policy is a textual string consisting of various mechanisms, each of which is either:

1. a literal block of one or more IP addresses
2. a hint that can be translated to one or more IP addresses using the Domain Name System (DNS)
3. a recursive SPF policy to replace or extend the current policy

The collection of IP addresses yielded by the policy constitute those explicitly authorized to send mail for a domain.

SPF: Sender Policy Framework (RFC 7208)

Example:

```
foo.com: v=spf1 ip4:192.0.2.1 a:bar.foo.com  
include:foo.net -all
```

Where to publish? DNS!

An SPF record is a DNS record that declares which hosts are authorized to send emails on behalf of a domain.

You use the TXT record for the sender domain.

When a receiving MTA receives an email from user@foo.com, it looks up the SPF policy for foo.com via a DNS query for foo.com type TXT.

DKIM: DomainKeys Identified Mail (RFC 6376)

Idea:

The sending MTA inserts a cryptographic signature which can only be validated with the public key associated with the sender domain

DKIM signature is included in a header in the email message

Where to publish the public key to validate signature? DNS!

Special record of type TXT. DKIM record names follow this format:

```
[selector]._domainkey.[domain]
```

The DKIM header includes a reference to the domain name where that TXT record can be found—a subdomain of the sender domain.

DMARC: Domain-based Message Authentication, Reporting, and Conformance (RFC 7489)

DMARC requires that either SPF or DKIM pass.

It gives a sender domain the ability to specify what an MTA should do with email that fails both. A DMARC policy also designates an email address to which violations should be sent.

Where to publish DMARC policy? DNS!

Special record of type TXT. DKIM record names follow this format:

```
_dmarc.[domain]
```

Summary

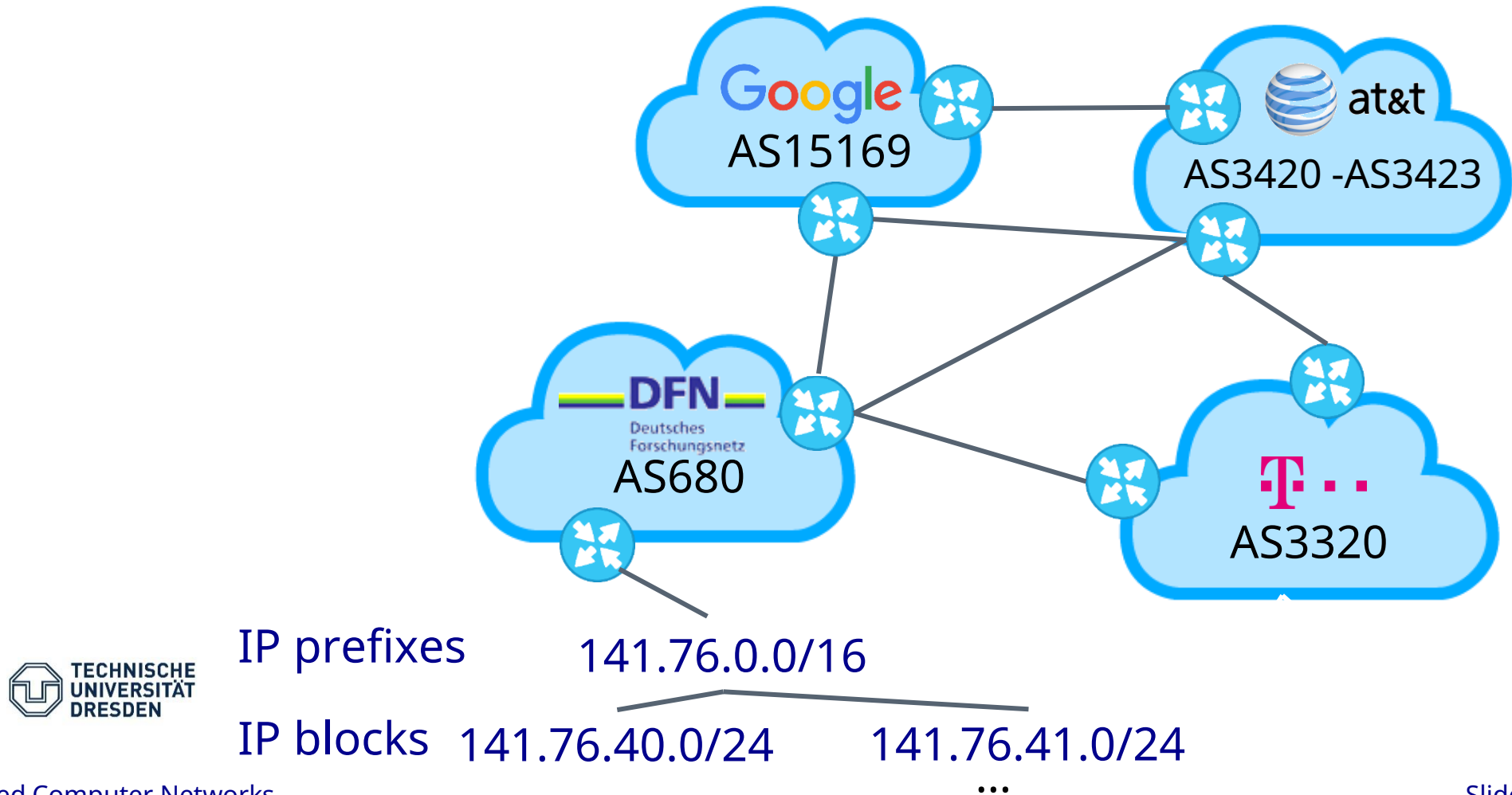
Operating a distributed, multi-stakeholder system securely is challenging

DNS is base even for security mechanisms but it needs to be secure before.

What happens this lecture?

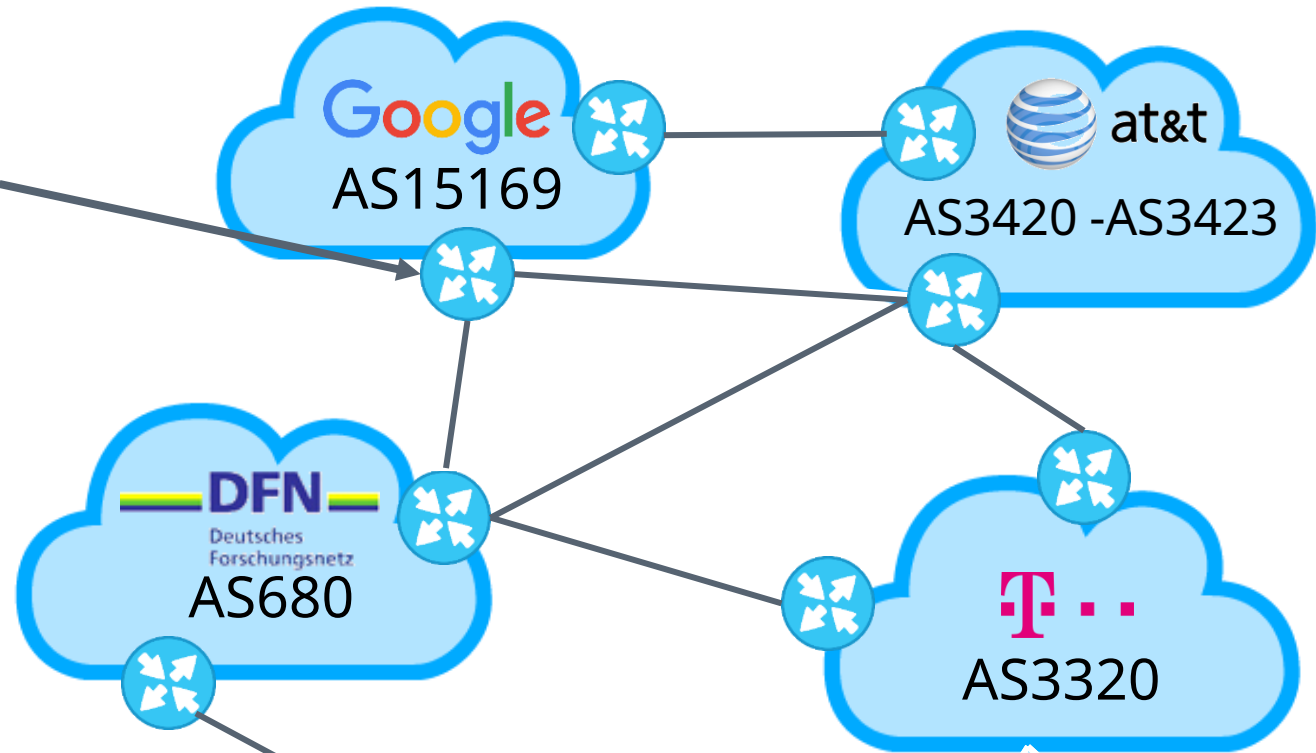
Inter-domain Routing Security

Autonomous systems = One domain



Autonomous systems = One domain

141.76.0.0/16 via DFN



IP prefixes

141.76.0.0/16

IP blocks

141.76.40.0/24

141.76.41.0/24

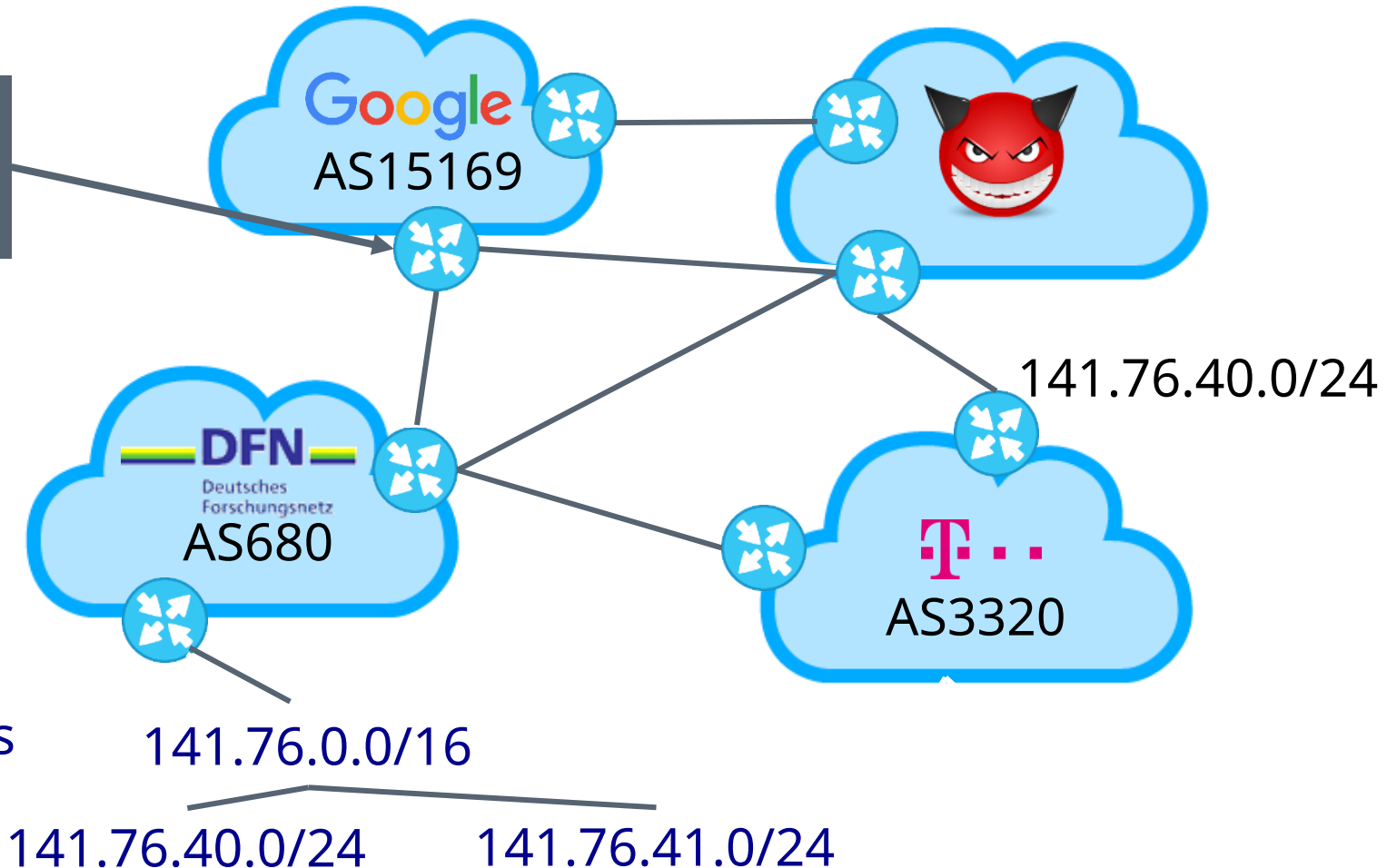
Prefix (or BGP) hijacking

An autonomous system attracts traffic to an IP prefix by incorrectly announcing reachability of prefixes.

Hijacking of address space

There are several options

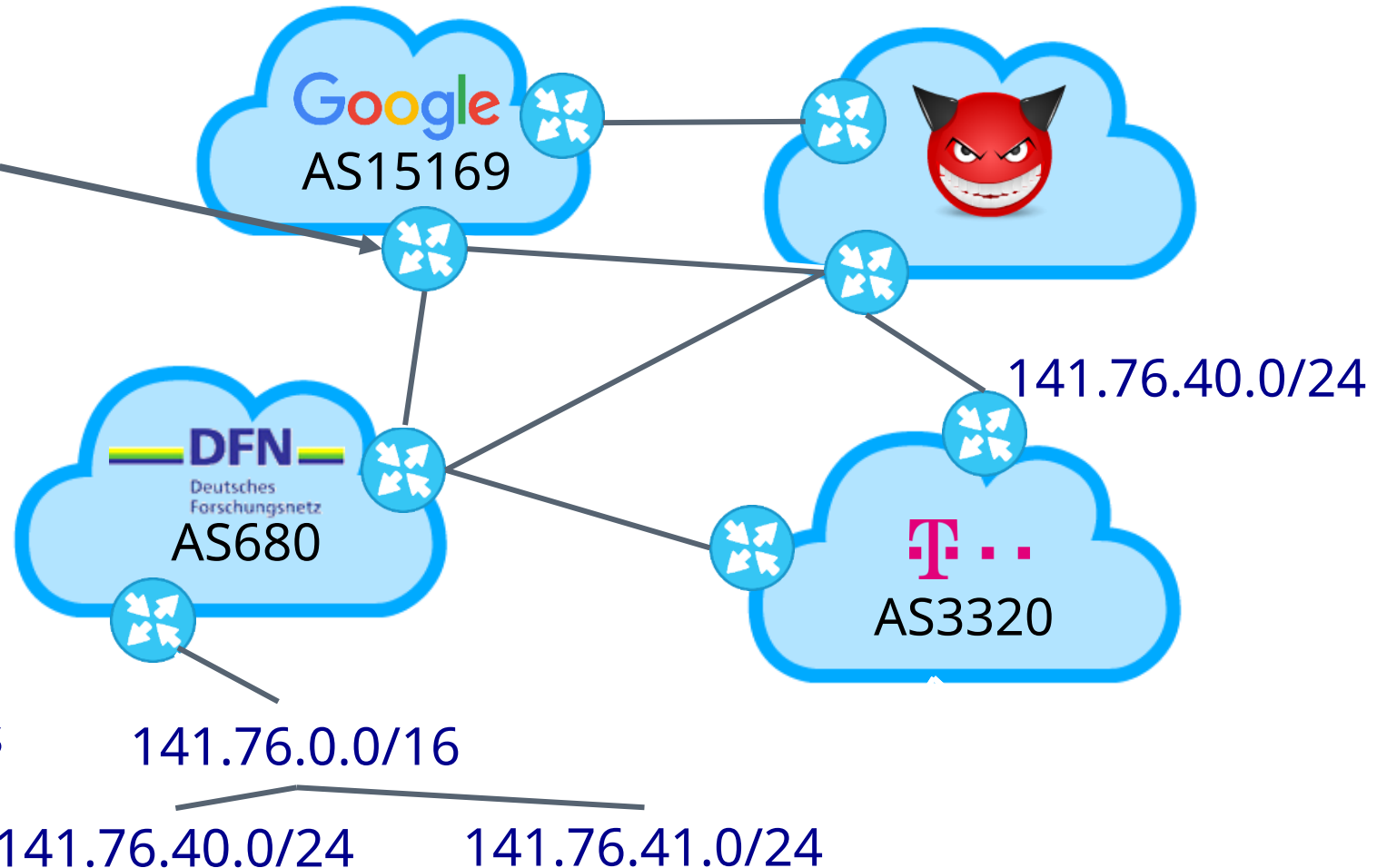
141.76.0.0/16 via



Hijacking of address space

There are several options

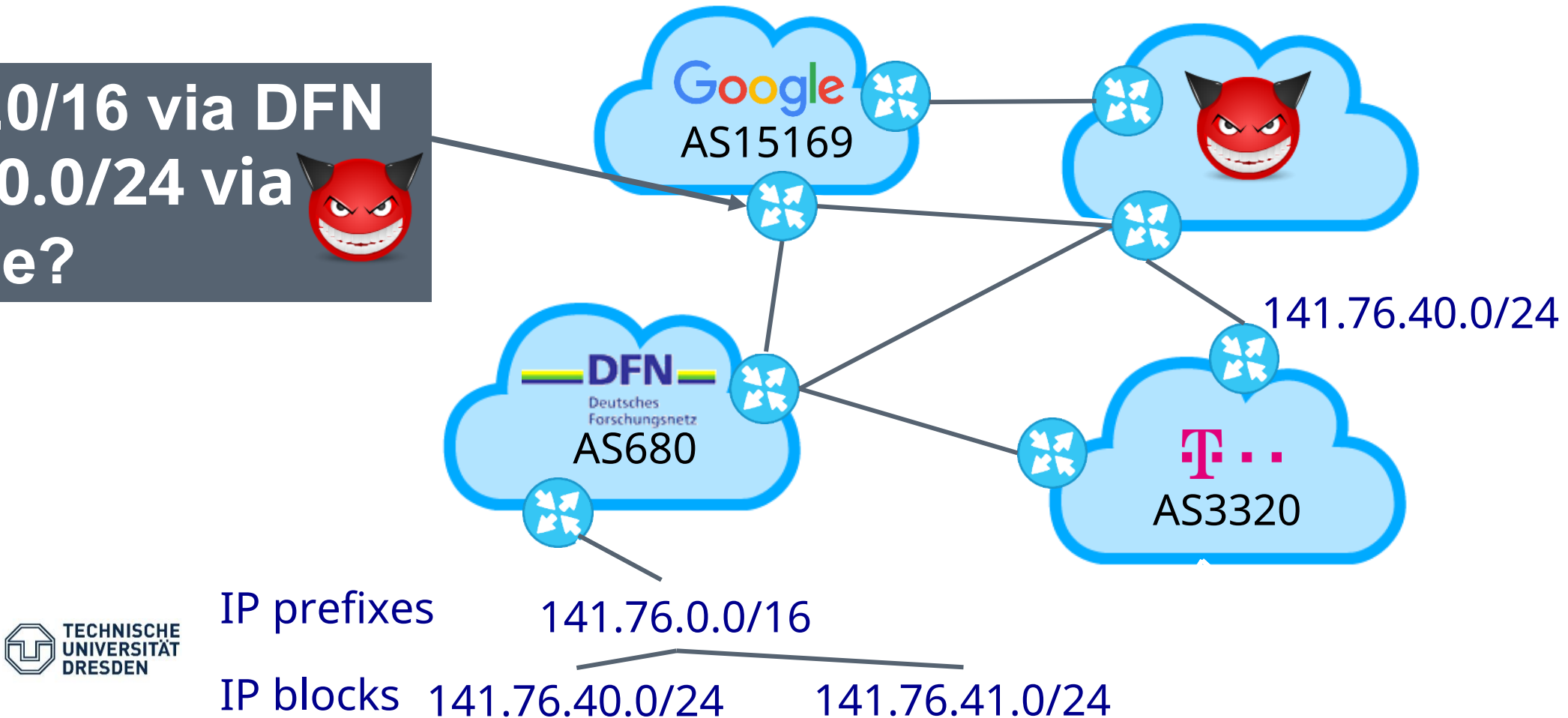
141.76.0.0/16 via DFN
141.76.40.0/24 via



Hijacking of address space

There are several options

141.75.0.0/16 via DFN
141.76.40.0/24 via 
What else?



Hijacking of address space

When the core protocol for Internet routing (BGP) was designed security was not considered.



IP prefixes

141.76.0.0/16

IP blocks

141.76.40.0/24

141.76.41.0/24

Summary of potential hijacking and route leak options

Origin hijack

Sub-prefix
hijack

Path
manipulations

Policy
violations

Problem

Original Design Choice in BGP (RFC 4271)

BGP is based on trust between peers

Implications

Any BGP speaker can claim to own an IP prefix

Any BGP speaker can modify the AS path

Receiver of a BGP update cannot verify the correctness of the data

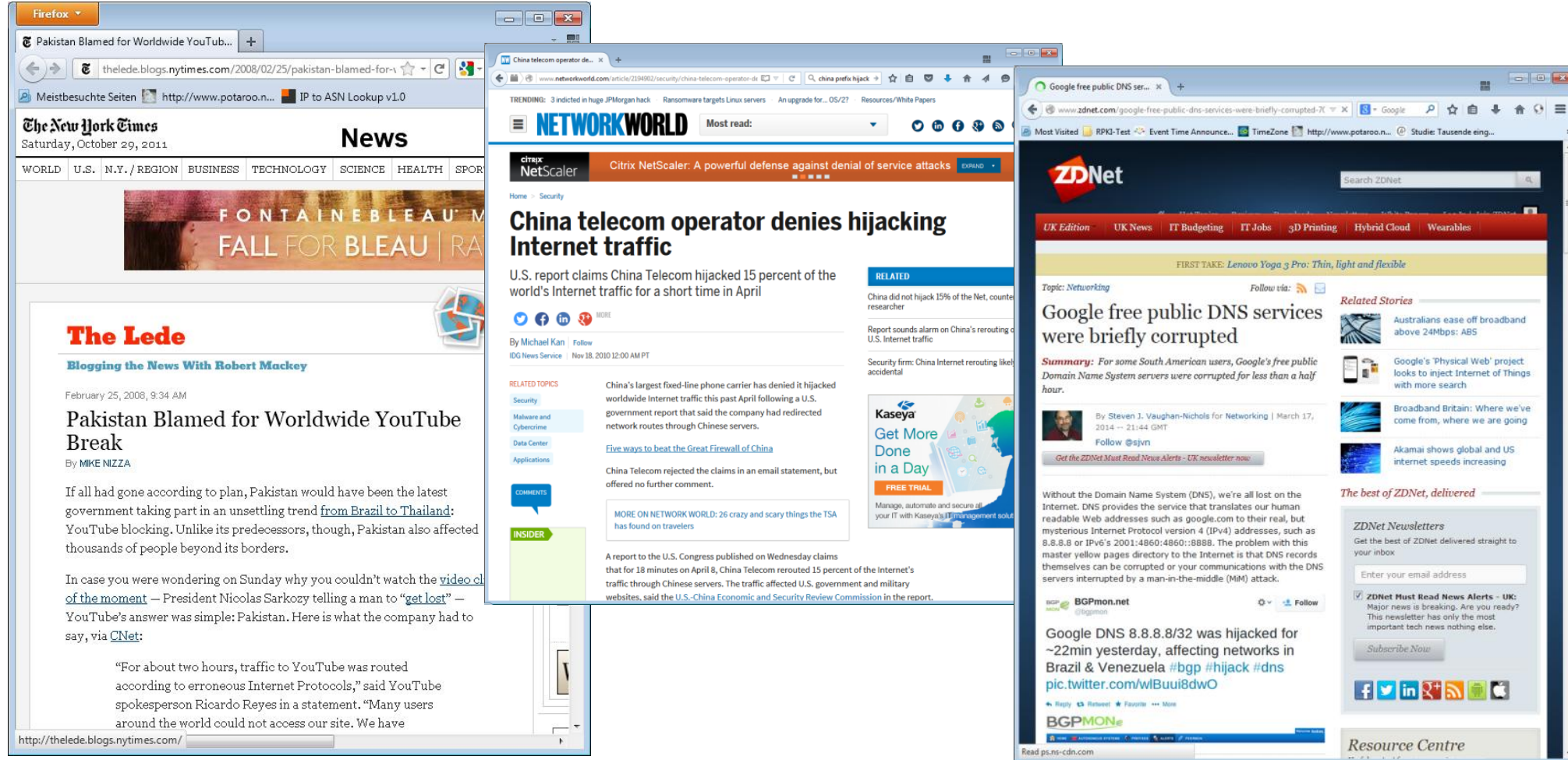
Compromise

Filtering

Considering data of the Internet Routing Registry

⇒ This is not enough anymore!

It's not fantasy.



Caveat: Reasons may also be misconfiguration ;-)

Political policies

THE WHITE HOUSE



JANUARY 16, 2025

Executive Order on Strengthening and Promoting Innovation in the Nation's Cybersecurity



BRIEFING ROOM

PRESIDENTIAL ACTIONS

MENU



(b) The security of Internet traffic depends on data being correctly routed and delivered to the intended recipient network. Routing information originated and propagated across the Internet, utilizing the Border Gateway Protocol (BGP), is vulnerable to attack and misconfiguration.

(i) Within 90 days of the date of this order, FCEB agencies shall take steps to ensure that all of their assigned Internet number resources (Internet Protocol (IP) address blocks and Autonomous System Numbers) are covered by a Registration Services Agreement with the American Registry for Internet Numbers or another appropriate regional Internet registry. Thereafter, FCEB agencies shall annually review and update in their regional Internet registry accounts organizational identifiers related to assigned number resources such as organization names, points of contact, and associated email addresses.

Katalog von Sicherheitsanforderungen

(ENTWURFSFASSUNG)

für das Betreiben von

Telekommunikations- und

Datenverarbeitungssystemen

sowie für die Verarbeitung

personenbezogener Daten nach

§ 167 Telekommunikationsgesetz (TKG)

Stand: Oktober 2025

ENTWURF

3.2 Bereitstellung von Routing-Informationen

Um falsche Routing-Informationen zu filtern (siehe Anhang, Kapitel 3.3), müssen die eigenen Routing-Informationen dokumentiert werden.

a)

- authentifizierte und autorisierte Routing-Informationen müssen in standardisierter Form bereitgestellt werden (beispielsweise mittels ROAs in RPKI)

b)

- Für diesen Punkt sind keine darüberhinausgehenden Maßnahmen verpflichtend

c)

- Für diesen Punkt sind keine darüberhinausgehenden Maßnahmen verpflichtend

3.3 Verhinderung der Weitergabe falscher Routing-Informationen

Die Verbreitung falscher Routing-Informationen kann zu Störungen führen und Angriffe begünstigen.

118 | I C ANHANG - TECHNISCHE SPEZIFIKATION FÜR PAKETVERMITTELTE NETZE ZUM KATALOG DER SICHERHEITSANFORDERUNGEN

a)

- Es muss sichergestellt werden, dass ausschließlich diejenigen IP-Präfixe originär in BGP annonciert werden, für die sie als Routen-Ursprung autorisiert sind
- Ein- und ausgehende Routing Annoncierungen der BGP-Nachbarn (außer die Annoncierungen der Transit-Provider) müssen gegen eine authentifizierte Quelle (beispielsweise RPKI) geprüft und gefiltert werden [MANRS] [RFC7454]

b)

- Für diesen Punkt sind keine darüberhinausgehenden Maßnahmen verpflichtend

c)

- Für diesen Punkt sind keine darüberhinausgehenden Maßnahmen verpflichtend

How can we solve this problem?

Problems:

No proof of address ownership.

No verifiable insights into AS relations.

Solution: Resource Public Key Infrastructure

Attestation for Internet resources.

Protection concepts proposed by the IETF

Route Origin Validation

ROV
(RFC 6811)

Mapping of IP prefixes to origin AS necessary
Including cryptographic proof
Prefix owner should be able to authenticate *Origin AS(es)*

Path Validation

BGPsec
(RFC 8205)

BGP path information are cryptographically secured
Paths will be signed

Policy Validation

ASPA
(draft-ietf-sidrops-aspa-verification)

Binds provider ASes to a customer AS

Challenges in protecting BGP

Cryptographic operations are complex

Prevent additional computational load on routers

Prevent additional memory requirements on routers

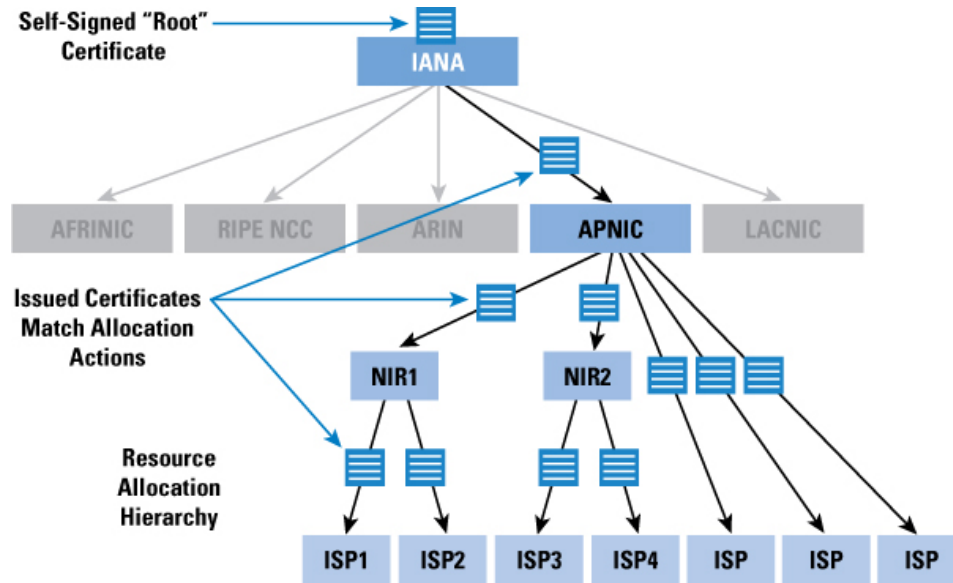
... If possible

Changing BGP core routers is not trivial

Companies have procurement cycles

Operators need the expertise about the new technology

The Resource Public Key Infrastructure (RPKI) is the basic solution proposed in the IETF



Source: Cisco

RPKI is a system that allows to attest the usage of IP addresses and ASNs (i.e., Internet resources) and thus includes cryptographically provable certificates.

Certificate hierarchy reflects IP-/AS-allocation in the Internet

Currently, each RIR creates a self-signed root certificate

Implementation of the RPKI started January 2011.

All RIRs participate!

Currently, we lack a single root of trust. There are five trust anchors.

Routing Origination Authorization (ROA) legitimates an AS to originate IP prefix(es)

Content of an ROA

Set of IP prefixes with minimal and maximal (optional) length

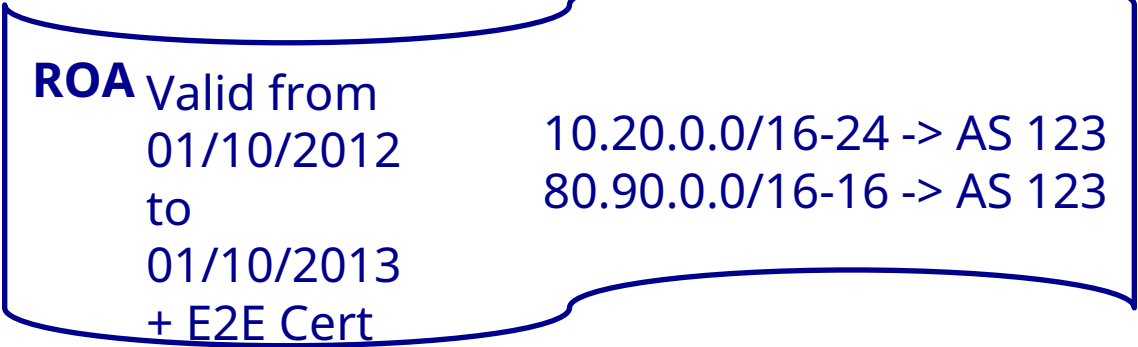
An AS number allowed to announce the prefixes

End-Entity-Certificate

ROA will be signed with the certificate of the RPKI

Note: Multiple ROAs per IP prefix possible

Example: AS 123 is allowed to announce network range 10.20.0.0/16 to 10.20.0.0/24 and 80.90.0.0/16 from 1st Oct. 2012 until 1st Oct. 2013



ROA Valid from
01/10/2012 10.20.0.0/16-24 -> AS 123
to 80.90.0.0/16-16 -> AS 123
01/10/2013
+ E2E Cert

A ROA is not a certificate



Certificate

Proofs **ownership** of an
IP prefix or an AS number



ROA

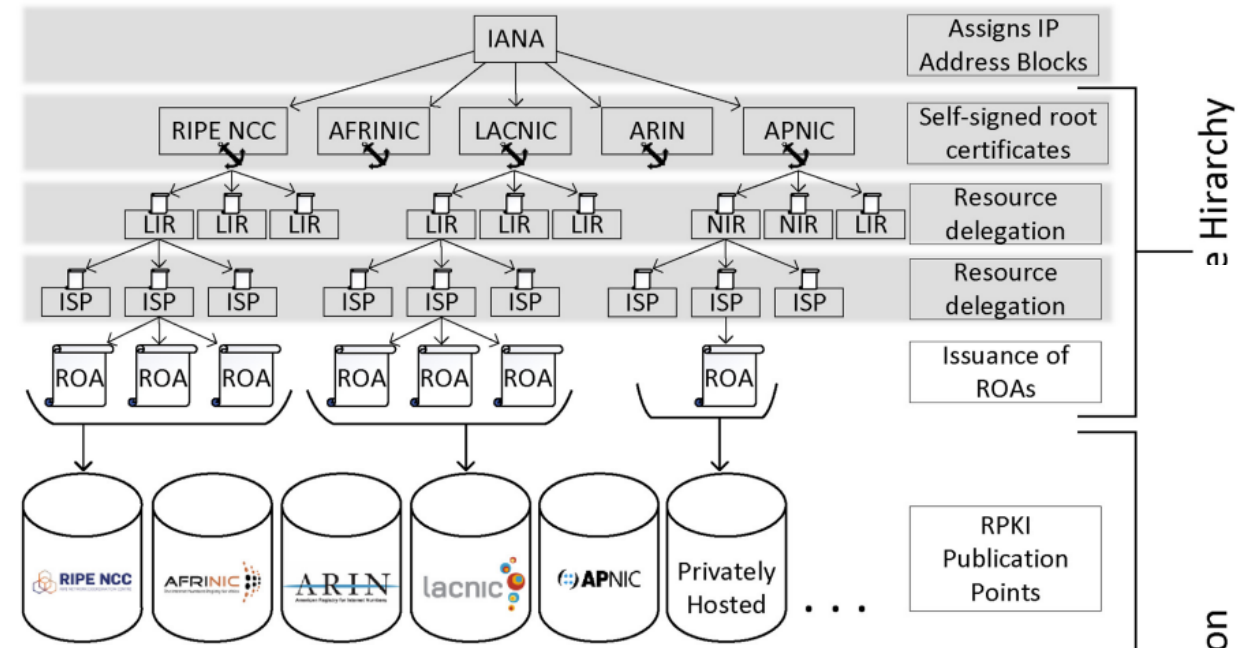
Proofs that an AS is **allowed
to announce** IP prefix(es)

RPKI architecture

There are two options to operate an RPKI repository

Hosted model, all certificates and ROAs are stored and managed by a RIR.

Delegated model, maintenance of the RPKI is delegated to third parties. Owners of Internet resources can run their own Certificate Authority (CA) to manage certificates and ROAs of IP prefixes. Running a publication point is also possible but not required.



RIPE and APNIC support both models.

RPKI MIRO allows for monitoring and inspection of RPKI objects

Originally developed by us but not maintained anymore

The screenshot displays the RPKI Repository Browser interface. The top navigation bar includes links for 'RPKI Repository Browser', 'RPKI Browser', 'Statistics', and 'About'. Below this, a secondary bar contains 'Filter', 'Clear Filter', 'List View', and 'RIPE'. The main content area is divided into three sections: a left sidebar with a tree view of certificates, a central pane showing the details of a selected certificate, and a right pane listing resources.

Left Sidebar (Certificate List):

- ripencc-ta.cer
 - a4250f3c598917bc119f7ddd423595bb7251181c.cer
 - 102ed0852ea4b4700eae91a42e9f7e0fe53497e6.cer
 - b1314c89c8803a127d62c83f01292a0474b9e6.cer** (selected)
 - OHOW3FTWhiq-VD3lwjGDiqc92Vs.cer
 - uhdYw1u0e_vOOpZZkPsfZmva8.cer

Filter Options Dialog:

- Choose filter attribute:** Filename, Location, Subject, Issuer, Serial Nr., Resource
- Search for:** [Text input field]
- Select file type:** All (selected), Only .cer, Only .roa
- Select validation status:** Passed (checked), Warning, Error
- Buttons: Clear, Apply Filter, OK

Central Pane (Certificate Details):

- Validation Status:** PASSED
- Validity Period:** 2015-04-22T12:29:22.000Z - 2020-04-22T12:29:22.000Z
- Errors:** None
- Warnings:** None
- subject:** CN=b1314c89c8803a127d62c83f01292a0474b9e6
- iKI:** B1314C89C8803A127D62C83F01292A0474B9E6
- Issuer:** CN=ripencc-ta
- iKI:** E8552B1FD6D1A4F7E404C6D8E5680D1EBC163FC3
- SerialNr:** 122
- Filename:** b1314c89c8803a127d62c83f01292a0474b9e6.cer
- Location:** rsync://rpki.ripe.net/repository/
- isE:** false
- isA:** true
- isA:** false
- Public Key:** Sun RSA public key, 2048 bits
modulus: 19138882686787109638459123509056288006693111056614826
7689467231031728942254738963663083307059847487556318262325584418
3493002787504972781180734224403831970613522148206912120764845683
4210659903809596054613933854422590279954157475636481232870206757
9075650274774795394961405961912534695789886043088504861342996695
8904428771612995116477927937673541238598672599596640467282174602
6283661633705428850245650752892991124742489726391820483183724446
0480856757526482414251569336827131357224820467753884396734910268
9701135877437971414366669465455558431385045981696609273422939720
8147961247719470085239285174427532861248550370260317
public exponent: 65537
- [ASCII download](#)

Right Pane (Resources):

Resources
24.132.0.0/14
45.8.0.0/13
45.80.0.0/12
45.128.0.0/11
128.0.0.0/16
128.7.0.0/16
128.16.0.0/16
128.39.0.0-128.41.255.255
128.45.0.0/16
128.65.0.0-128.79.255.255
128.86.0.0/15
128.93.0.0/16
128.98.0.0/16
128.124.0.0/16
128.127.0.0/16
128.130.0.0/15
128.139.0.0-128.142.255.255
128.176.0.0/16
128.178.0.0/15
128.199.0.0/16
128.204.0.0/16

Cloudflare provides a service similar to RPKI MIRO

<https://rpki.cloudflare.com/?view=explorer>

Resource List

Hierarchical View

Found 217 ROAs and 10 certificates

ROAs

ASN	Prefix	Max Length	IP Family	Trust Anchor	Emitted	Expiration
AS680	134.28.0.0/16	/16	IPv4	RIPE	2.1.2024	in 7 months
AS680	139.18.0.0/16	/16	IPv4	RIPE	1.1.2024	in 7 months
AS680	193.22.32.0/20	/20	IPv4	RIPE	1.1.2024	in 7 months
AS680	193.22.48.0/23	/23	IPv4	RIPE	1.1.2024	in 7 months
AS680	141.14.0.0/16	/16	IPv4	RIPE	2.1.2024	in 7 months

Prefix: 141.14.0.0/16
Max Length: /16
ASN: 680
Emitted: Tue, 02 Jan 2024 12:32:51 GMT
Validity: Tue, 02 Jan 2024 12:32:51 GMT - Tue, 01 Jul 2025 00:00:00 GMT
Trust Anchor: RIPE
Name: 0c15c2ca09073d19e0990ae1ad03118d717d88f0
Key: 0c15c2ca09073d19e0990ae1ad03118d717d88f0
Parent Key: 6857a9bb297f0b3c444ab0af673a52750e979b4e
Path: rsync://rpki.ripe.net/repository/DEFAULT/d4/9d0a15-d546-4970-9a81-8596c2af9dcc/1/DBXCygmHPRngmQhrQMRjXF9iPA.roa

Trust AnchorCertificateROA fileROASelected

RIPE
Trust Anchor

ec334d0f3a18be0021b7b8e842287641acb1731a
1 ROAS

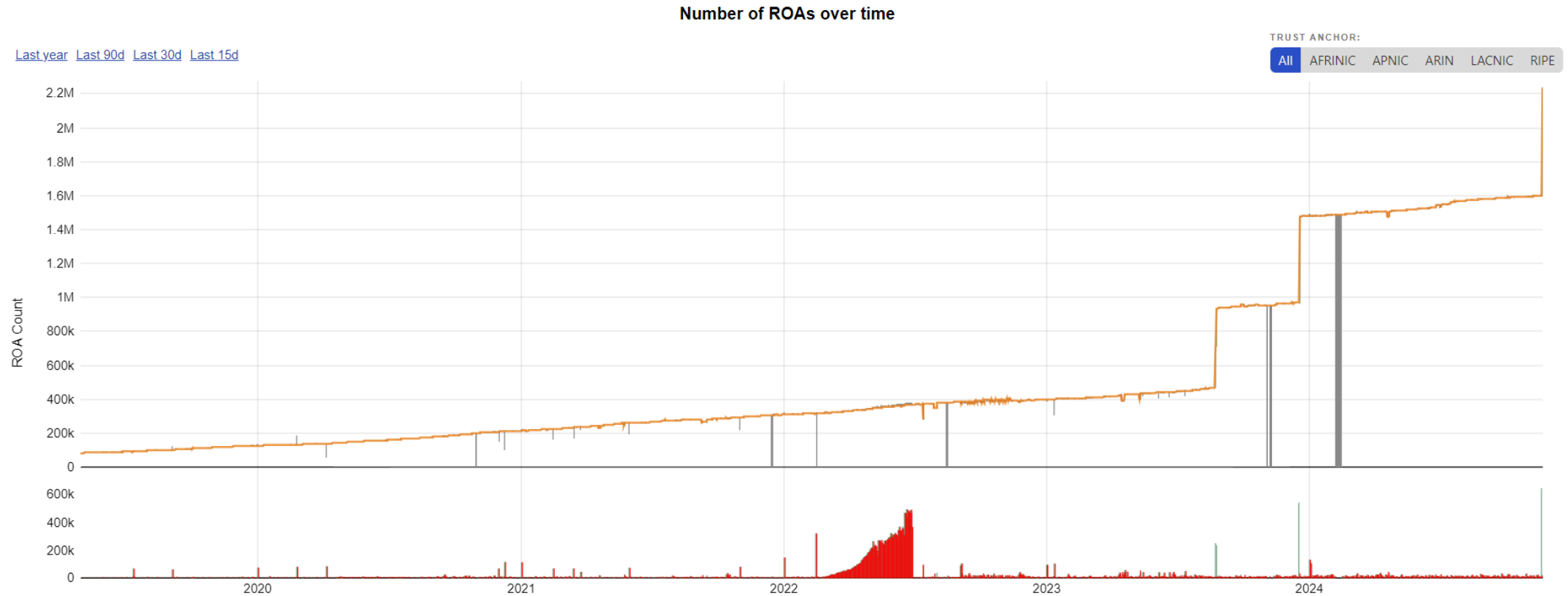
2a94a8dd554ae701072099c70b6407555ddde669
1 ROAS

6857a9bb297f0b3c444ab0af673a52750e979b4e
1 ROAS

0c15c2ca09073d19e0990ae1ad03118d717d88f0
AS680

141.14.0.0/16
AS680

Creation of ROAs over time



Prefix origin validation using RPKI consists in two steps

(1) Validation of
ROAs



(2) Validation of
BGP Updates

Performed at external cache

Performed at BGP router
No additional cryptographic
operations necessary

Prefix origin validation using RPKI consists in two steps

(1) Validation of
ROAs



(2) Validation of
BGP Updates

Performed at external cache

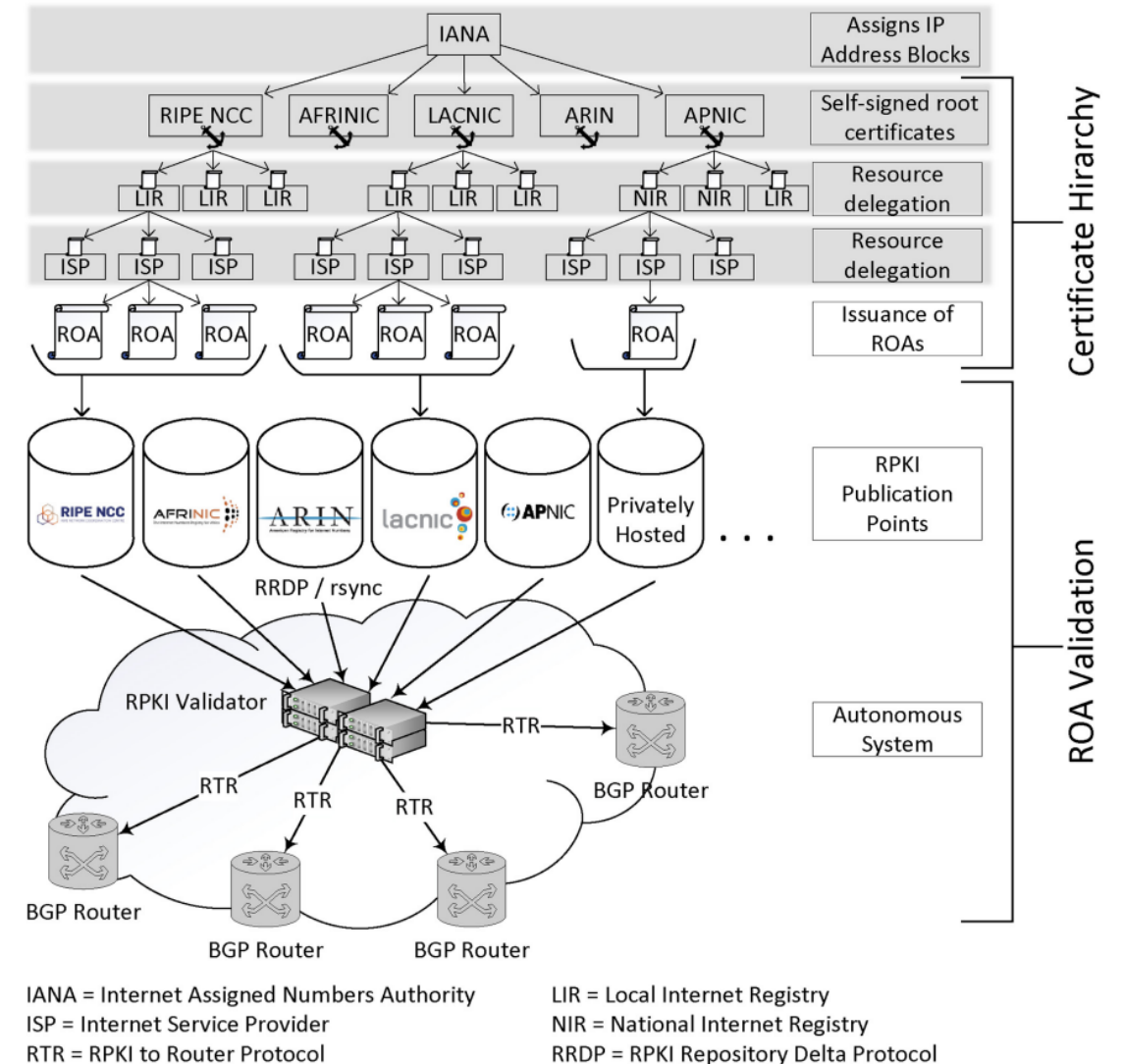
Performed at BGP router
No additional cryptographic
operations necessary

How does the RPKI data comes to the BGP router?

RPKI and ROV Architecture

RPKI Relying Parties (or cache servers) fetch RPKI objects and perform the cryptographic validation.

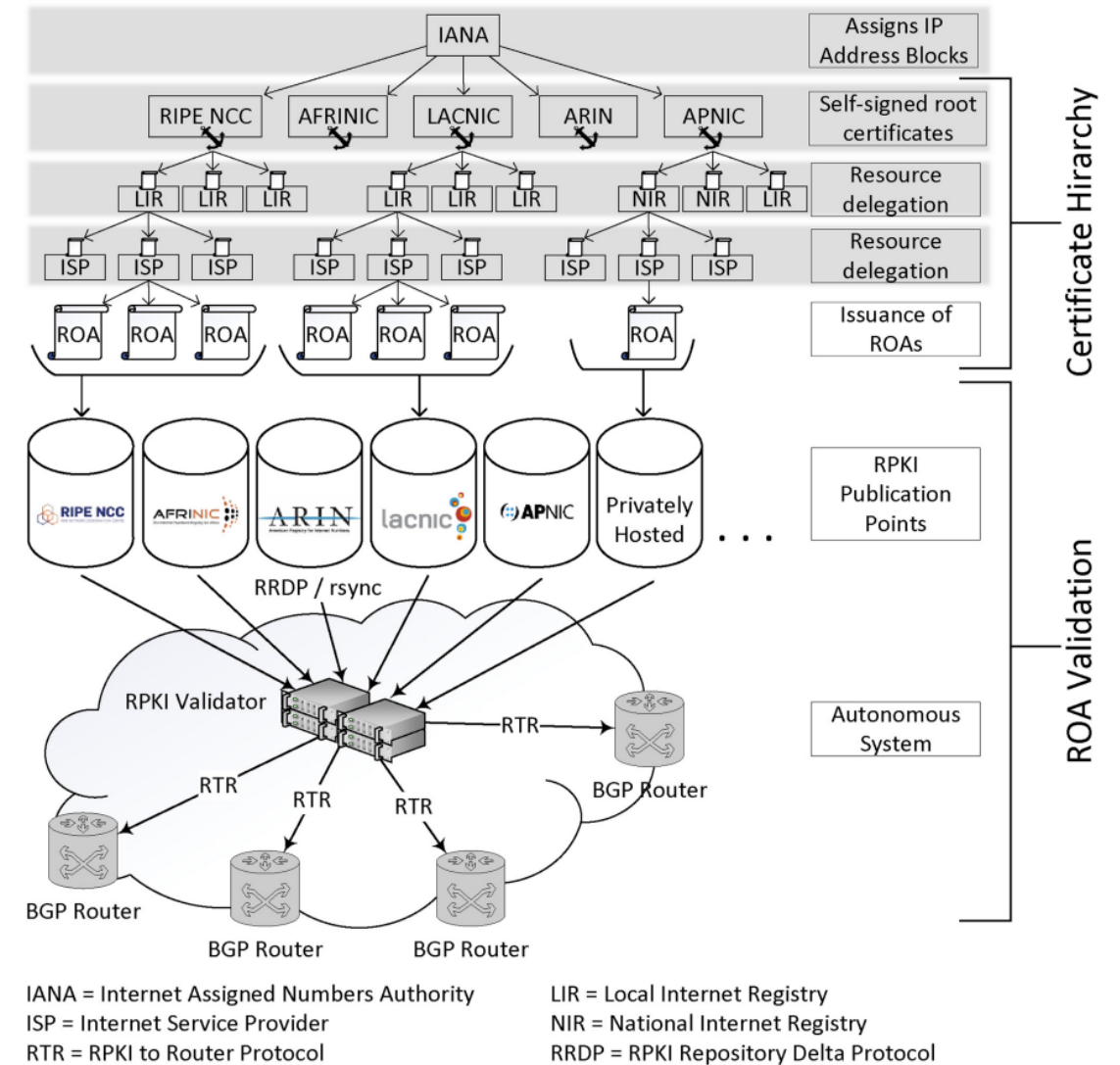
They provide cryptographically valid ROAs to routers, e.g., using the RPKI RTR protocol.



RPKI and ROV Architecture

Operational challenge?

What happens if publication point is staled?



RPKI relying party software

Name	Initial Release Year	Language	Maintained
RPKI Validator 1 [67]	2011	Scala	✗
Rcynic [59]	2012	C/Python	✗
RPSTIR [68]	2012	C	✗
OctoRPKI [69]	2019	Go	✓
FORT-Validator [70]	2019	C	✓
rpki-client [71]	2019	C	✓
Routinator 3000 [72]	2019	Rust	✓
RPSTIR2 [73]	2020	Go	✓
rpki-prover [74]	2020	Haskell	✓
RPKI-Validator 3 [58]	2021	Java	✗

RTRlib is reference implementation of RPKI RTR client in C. An open-source project co-founded by NETD

General objective

Implementation of the RPKI-RTR client protocol in C

Details

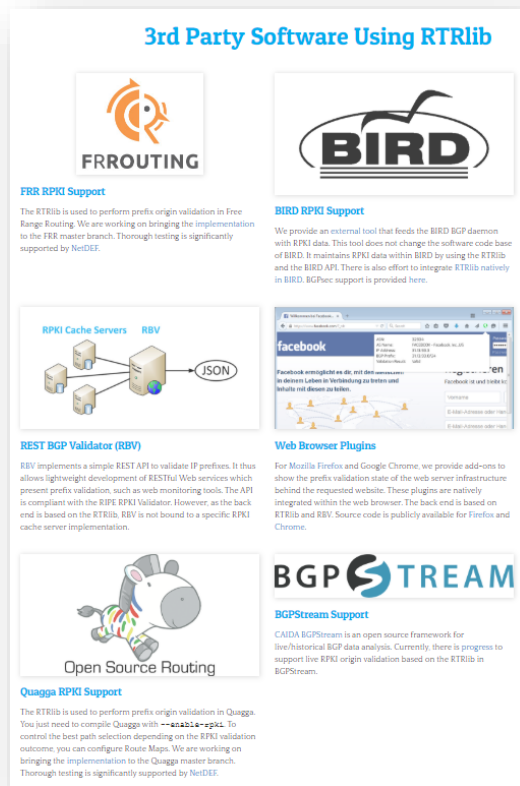
Fetch validated prefixes + origin ASes from RPKI cache

Keep the routers validation database in sync

Provide an interface between local database and routing daemon to access validated objects

Allow also for validation of BGP updates

Conforms to relevant IETF RFCs/drafts



ROAs and ROV

How do you validate a route in principle?

Check if AS is allowed to announce IP prefix = check the corresponding ROA

Consider all valid ROAs

Find those ROAs that cover the announced prefix based on classless-inter domain routing (CIDR)

Compare covering ROAs with route announcement

There are three states: VALID, INVALID, NOT_FOUND

RPKI implements a positive attestation

If a ROA for a prefix exists, announcements of all origin ASes that are not included or exceed the prefix

ROAs and ROV. How do you validate a route in principle?

Valid ROAs

10.20.0.0/16-17 {AS10,
AS20}
10.20.30.0/24-24 {AS10}

BGP announcements

10.20.0.0/16: AS20

10.20.1.0/24: AS10
10.20.0.0/16: AS100

1.2.3.0/24: AS10

**Check if AS is allowed to announce IP prefix =
check the corresponding ROA**

Consider all valid ROAs

Find those ROAs that cover the announced prefix
based on classless-inter domain routing (CIDR)

Compare covering ROAs with route announcement

There are three states: VALID, INVALID, NOT_FOUND

RPKI implements a positive attestation

If a ROA for a prefix exists, announcements of all
origin ASes that are not included or exceed the prefix
length will be considered INVALID.

ROAs and ROV. How do you validate a route in principle?

Valid ROAs

10.20.0.0/16-17 {AS10,
AS20}
10.20.30.0/24-24 {AS10}

BGP announcements

10.20.0.0/16: AS20 VALID

10.20.1.0/24: AS10 INVALID

10.20.0.0/16: AS100 INVALID

1.2.3.0/24: AS10 NOT FOUND

Check if AS is allowed to announce IP prefix = check the corresponding ROA

Consider all valid ROAs

Find those ROAs that cover the announced prefix based on classless-inter domain routing (CIDR)

Compare covering ROAs with route announcement

There are three states: VALID, INVALID, NOT_FOUND

RPKI implements a positive attestation

If a ROA for a prefix exists, announcements of all origin ASes that are not included or exceed the prefix length will be considered INVALID.

Some common pitfalls (examples)

Both announcements are invalid if not ROA exists

Case 1: Missing Customer (or Sibling) Legitimation

ROA created: 12.0.0.0/8-9 -> AS 7018

AS 27487 announces 12.0.19.0/24

AS 2386 announces 12.1.216.0/24

=> Consider sub-allocations, start most specific

Case 2: (De-)Aggregation

ROA created: 78.192.0.0/10-10 -> AS 12322

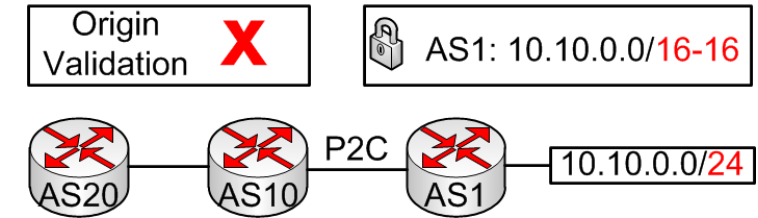
Usual announcement: 78.192.0.0/10

For 30 minutes: 78.192.10.0/24 ...

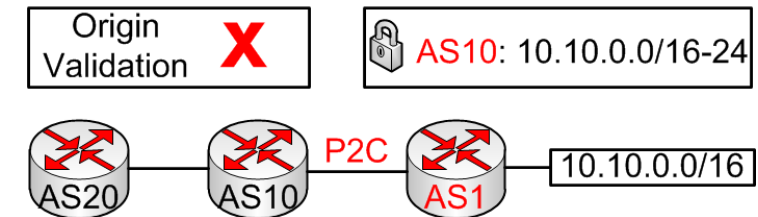
=> Configure the max ROA prefix length explicitly

Classification of potential ROA misconfigurations

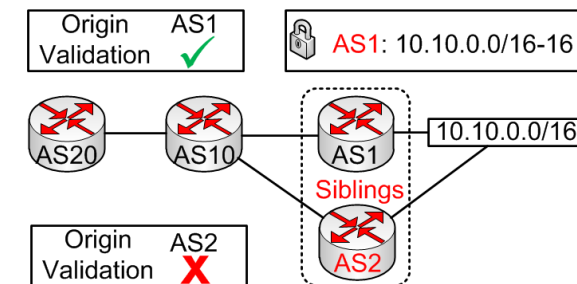
1. Valid origin, announced prefix is more specific



2. Provider does not consider customer

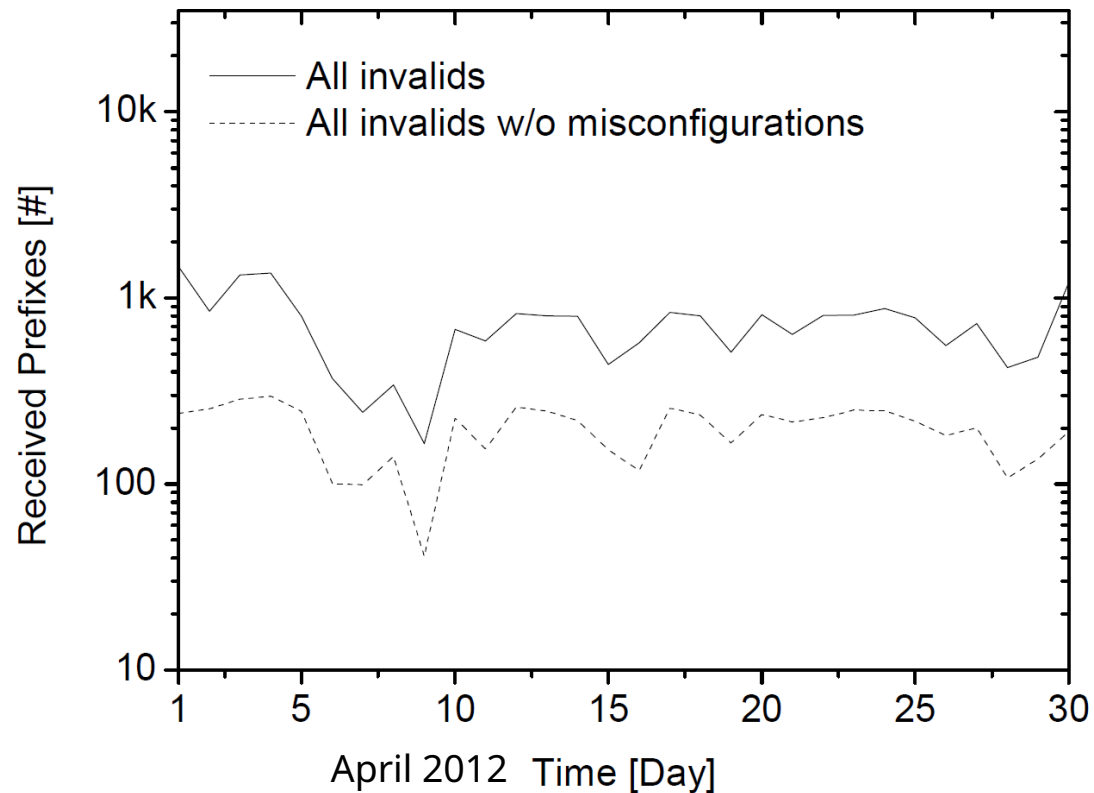


3. Additional ASes of a company are not authorized



Unexpected deployment behavior

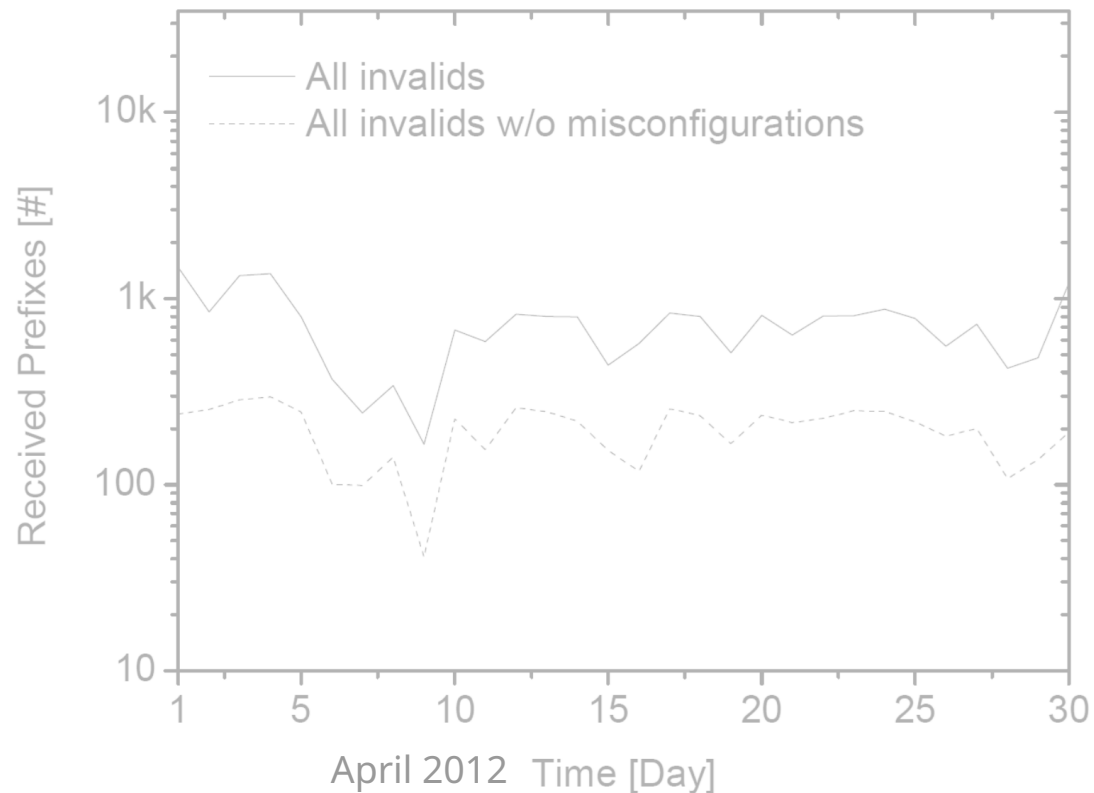
[ACM CCR, 42(4), 2012]



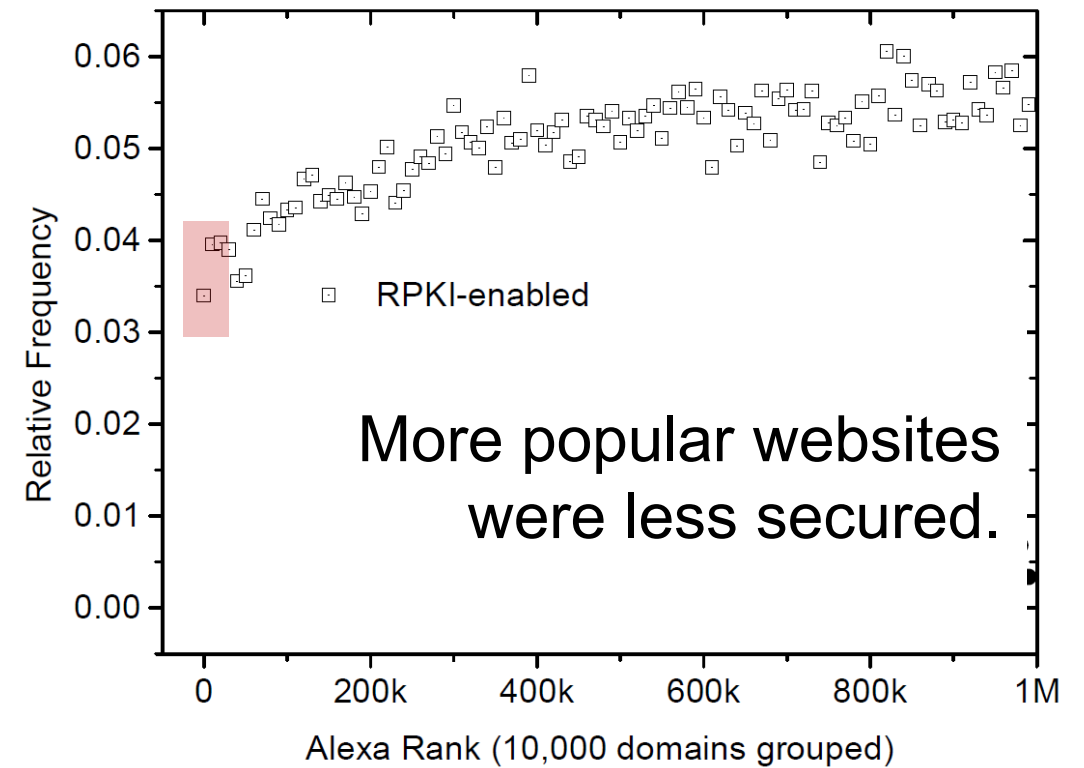
We developed a method to identify common errors, interacted with operators, and improved the situation.

Unexpected deployment behavior

[ACM CCR, 42(4), 2012]



[ACM HotNets 2015]



A router still needs to map ROV validation outcome to BGP policies

You do want to filter RPKI Invalids.

You do want to accept RPKI Valids.

What do you do with Not Found?

Example configuration on Cisco router, deploying different preferences depending on the ROV outcome

Route map named rtmap-PEX1-3 sets a local preference of 50 for invalid prefix/AS pairs, 100 for not-found prefix/AS pairs, and 200 for valid prefix/AS pairs.

The local preference values are set for incoming routes from the neighbor at 10.0.102.1.

The neighbor at 10.0.102.1 is an eBGP peer.

```
router bgp 65000
  address-family ipv4 unicast
  neighbor 10.0.102.1 route-map rtmap-PEX1-3 in
  bgp bestpath prefix-validate allow-invalid
  !
  route-map rtmap-PEX1-3 permit 10
    match rpki invalid
    set local-preference 50
  !
  route-map rtmap-PEX1-3 permit 20
    match rpki not-found
    set local-preference 100
  !

  route-map rtmap-PEX1-3 permit 30
    match rpki valid
    set local-preference 200
  !

  route-map rtmap-PEX1-3 permit 40
```

Many Internet Exchange Points (IXPs) support ROV on their route servers

The screenshot displays the BCIX Route Server interface. The left sidebar lists route servers: rs01.bcix.de (IPv4), rs01.bcix.de (IPv6), rs02.bcix.de (IPv4), and rs02.bcix.de (IPv6). The main content area shows the URL `https://lg.bcix.de/routeservers/rs01-bcix-v4/neighbors/pb_0021_as13237/routes` and the title `rs01.bcix.de (IPv4) » AS13237 - euNetworks GmbH`. A search bar is present with the text "Filter by Network or Next-Hop". Below it, a table titled "ROUTES FILTERED" shows 30 routes. The table has columns for Network, Next-Hop, AS Path, Local Pref., MED, and Origin. The first six routes are highlighted with a star icon and have a note: "Origin AS not in peer AS-SET". The last route is highlighted with a star and a blue circle icon and has a note: "IRRDB Prefix not found in AS-SET". On the right side, there are status boxes for "Last Reboot" (Thursday, July 11, 2024 3:16 AM), "Last Reconfig" (Thursday, November 21, 2024 9:47 AM), and "Daemon is up and running". Below these is a "RELATED NEIGHBORS" section showing connections to rs01.bcix.de (IPv6) and rs02.bcix.de (IPv4).

Network	Next-Hop	AS Path	Local Pref.	MED	Origin
★ 144.36.136.0/23 Origin AS not in peer AS-SET	193.178.185.39	13237 3573 8220 4775 21433	100	0	IGP
★ 144.36.168.0/23 transit-free ASN in AS-Path	193.178.185.39	13237 3573 8220 3356 17639 21433	100	0	IGP
★ 144.36.228.0/23 Origin AS not in peer AS-SET	193.178.185.39	13237 3573 8220 9498 21433	100	0	IGP
★ 144.36.252.0/23 transit-free ASN in AS-Path	193.178.185.39	13237 3573 8220 3356 17639 21433	100	0	IGP
144.36.46.0/24 Origin AS not in peer AS-SET	193.178.185.39	13237 3573 8220 9299 21433	100	0	IGP
★ 147.189.40.0/24 IRRDB Prefix not found in AS-SET	193.178.185.39	13237 4749	100	0	Incomplete
★ 170.248.124.0/23 Origin AS not in peer AS-SET	193.178.185.39	13237 3573 8220 4775 21433	100	0	IGP

When we started RPKI analysis ...

2012 [ACM CCR'12], misconfigurations were common

Now, misconfigurations are negligible

2015 [ACM HotNets'15], no CDN was involved in RPKI

Now, Cloudflare, Google ... are pushing

2018 [ACM CCR'18], few ASes deployed RPKI filtering

Now, many ASes (e.g., AT&T) deploy RPKI filtering

Did you wonder why we don't have complete deployment, neither right from the beginning nor now?

Politics. Business decisions.

Reasons for not deploying RPKI

Political reasons

RIR are trust anchors

Local law may instruct RIR to revoke certificates

ROAs become invalid

Out of control of the operator

Business reasons

RPKI implements a positive attestation model

ISPs have to add prefix-AS relation in advance

Might conflict with business policies

Cost and complexity reasons